

Cost-Efficient Navigation and Parking Selection in Urban Networks

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Abstract—This project presents a solution for cost-efficient navigation and parking selection in urban networks. The objective is to guide a driver to an available parking garage near a desired destination while accounting for travel time, walking distance, garage availability, and parking price. Two approaches were implemented and compared: A* search and Q-learning. The A* approach used heuristic pathfinding to identify efficient routes and parking choices within the urban network, while the Q-learning approach learned routing and parking behavior through repeated interaction with the environment. Simulation results showed that Q-learning achieved lower total travel time, whereas A* produced lower parking cost and shorter walking distance. Furthermore, A* did not experience the simulation failures observed in the Q-learning method, indicating greater reliability under the tested conditions. Overall, the results show that Q-learning has strong potential for adaptive parking decision-making, while A* remains a simpler and more robust solution for this problem.

Index Terms— Smart parking, urban networks, dynamic parking assignment, A* search, Q-learning, path planning, intelligent transportation systems.

I. INTRODUCTION

Parking is a persistent challenge in urban transportation systems, especially in dense cities where drivers often spend substantial time searching for parking, which increases traffic congestion, fuel consumption, emissions, and travel uncertainty. As a result, parking has become an important problem in intelligent transportation systems, with applications in smart cities, navigation systems, and real time parking guidance.

The problem is not simply to find the closest available parking location. In practice, drivers must consider several competing factors, including travel time, walking distance to the destination, parking cost, and garage availability. Prior smart-parking research has shown that urban parking allocation should not be treated as a simple nearest-space problem, since coordinated assignment can outperform uncontrolled or guidance-only systems [1]. Broader survey work also shows that modern smart parking systems increasingly combine sensing, prediction, optimization, and intelligent control to better manage limited parking resources [2].

In this project, we address the problem of cost-efficient navigation and parking selection in an urban network. The objective is to guide a driver from a starting location to a parking garage near a destination while minimizing travel time, parking price, and walking distance, subject to garage capacity constraints. To solve this problem, we implemented and compared two approaches: A* search and Q-learning. The A* method uses heuristic pathfinding to identify efficient routes and parking choices, while the Q-learning method learns routing and parking behaviour through repeated interaction with the environment. Our results show that both approaches can produce effective parking decisions, while also revealing different trade-offs between deterministic search and adaptive learning in dynamic parking environments.

II. RELATED WORK

Smart parking has been studied from several perspectives, including optimization, reservation systems, dynamic allocation, pricing, and reinforcement learning. Geng and Cassandras proposed one of the foundational smart-parking frameworks, where parking is assigned and reserved according to an objective function combining proximity to destination and parking cost [1]. Their work solves a mixed-integer linear program at each decision point and demonstrates that coordinated allocation improves utilization relative to uncontrolled parking behaviour [1]. This work is important because it formalizes parking allocation as a decision problem rather than simple driver guidance.

A broader review by Fahim et al. surveys smart parking systems across sensing technologies, networking infrastructures, interfaces, and computational approaches [2]. Their review shows that smart parking is inherently interdisciplinary and that practical systems must integrate information acquisition with decision-making logic. A key limitation of many real deployments is that they emphasize detection and monitoring more than adaptive route or policy optimization [2].

More recent work has pushed toward dynamic and scalable urban parking assignment. Mladenovic et al. propose a scalable dynamic parking allocation framework capable of coordinating parking decisions at city scale and handling real-time updates [3]. Their approach shows that dynamic parking allocation is significantly harder than static assignment, especially when the system must repeatedly revise decisions over time (3). This supports the relevance of this project's

dynamic setting, where travel times and garage states change during navigation.

Reinforcement learning has also been applied to online parking assignment. Zhang et al. formulate online parking assignment under partial observation as a multi-agent deep reinforcement learning problem and compare it against first-come-first-served and forecasting-based baselines [4]. Their results show that reinforcement learning can better account for long-term effects under uncertainty [4]. Similarly, reinforcement-learning-based dynamic pricing work has shown that reinforcement learning is well suited to sequential parking decisions because it can adapt to changing demand and occupancy patterns over time [5].

This work differs from the literature in two ways. First, instead of focusing only on centralized optimization or only on pricing, it compares a classical path-planning method with a learning-based method within the same urban parking framework. Second, the implementation explicitly combines navigation and garage selection, so the driver must not only choose a good parking garage but also travel through a dynamic graph to reach it. This gives the project a practical artificial intelligence focus while remaining computationally manageable.

III. SYSTEM MODEL AND PROBLEM DESCRIPTION

Fig. 1 illustrates the system model used in this project. The DPAUA problem is modelled as a graph $G=(V,E)$, where each node represents a location in the road network and each edge represents a road connection between neighbouring intersections. In the implementation, the graph is stored using an adjacency-list structure. Each node contains a Python dictionary of attributes, including coordinates and node type. Most nodes represent ordinary road intersections, while garage nodes additionally store parking-related information such as current capacity, maximum capacity, and parking price. Each edge stores a destination node, a minimum travel time, and a current travel time that captures dynamic traffic conditions.

The figure also shows the interaction between the environment and the utility-based agent. The agent receives percepts including the current node, neighbouring roads with associated traffic conditions, and garage information such as price and available capacity. Based on this information, the agent selects actions such as moving to a neighbouring node, choosing a target garage, or stopping once a feasible garage is reached. In this way, the system combines navigation and parking selection within the same decision-making framework.

The main constraints of the problem are as follows. First, a garage is only considered feasible if its walking distance to the destination is below a maximum threshold, W_{max} . Second, a full garage should be avoided. Third, the decision should balance road travel time, parking cost, and walking distance after parking rather than optimizing only one factor. This creates a multi-criteria decision problem in which a cheaper garage may require more driving or walking, while a closer

garage may be more expensive or congested.

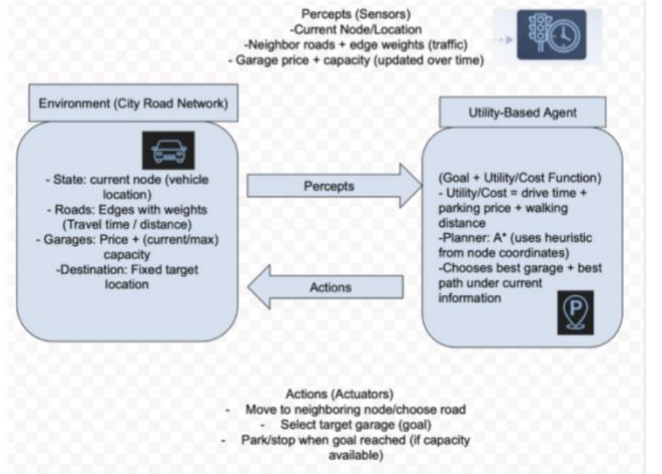


Fig. 1. System model for dynamic parking assignment in urban areas.



Fig. 2. an example instance of the graph-based urban parking environment, including the road network, garage locations, start node, destination, selected garage, and the route chosen by the planner.

An example of the simulated environment is shown in Fig. 2. The road network is represented as a grid-like graph in which ordinary intersections are connected by road segments and a subset of nodes correspond to parking garages. The figure also illustrates a sample routing outcome, including the start location, destination, selected garage, and chosen path. This example highlights the multi-criteria nature of the problem, since the selected garage must balance travel time, parking price, garage capacity, and walking distance to the destination.

IV. PROPOSED APPROACH

To solve the dynamic parking assignment problem, two approaches were implemented and compared: A* search and Q-learning. The A* approach provides a heuristic path-planning solution based on current graph information, while the Q-learning approach learns a parking and routing policy through repeated interaction with the environment. These methods were selected to compare a classical search-based technique with a learning-based method under the same graph-based urban parking framework.

A. A* Search Approach

The first approach uses A* search to compute a route from the current vehicle location to a selected parking garage. The implementation does not search over all garages simultaneously. Instead, it first evaluates candidate garages using a cost estimate, selects the most promising garage, and then runs A* to that garage.

Garage selection is performed by pruning garages whose walking distance to the destination exceeds the maximum allowed walking threshold $W_{\max}$. For each remaining garage, an estimated total cost is computed as

$$C_g = d(g, d) + d(s, g) + p_g$$

where $d(g, d)$ is the Manhattan distance from garage g to the destination, $d(s, g)$ is the Manhattan distance from the current state s to the garage, and p_g is the parking price of garage g . The garage with the minimum estimated cost is selected as the target garage.

After a target garage is chosen, A* search is used to compute a path from the current vehicle location to that garage. Each node n is evaluated using

$$f(n) = g(n) + h(n)$$

where $g(n)$ is the accumulated road travel time from the start node to node n , and $h(n)$ is a heuristic estimate of the remaining cost to the target garage. In this implementation, the heuristic is the Manhattan distance between the current node and the selected garage:

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

where (x_n, y_n) and (x_g, y_g) are the coordinates of node n and the target garage, respectively. The path cost $g(n)$ is computed using the current travel time stored on each edge, allowing the search to reflect dynamic traffic conditions.

The final route is not fixed for the entire trip. During navigation, the remaining path is monitored, and A* is re-run whenever the current remaining travel time becomes significantly worse than the previous estimate. In the implementation, replanning is triggered when the travel time of the remaining path exceeds 1.4 times the expected

remaining cost. This allows the agent to adapt to changing traffic conditions while still using a deterministic search procedure.

This approach was selected because it is simple, interpretable, and computationally efficient for graph-based routing. It also provides a strong baseline for comparing a classical search-based planner against a learning-based method.

B. Q-Learning Approach

The second approach uses Q-learning, a model-free reinforcement learning method, to learn routing and parking decisions through repeated simulation. Unlike A*, this method does not explicitly choose a garage first. Instead, it learns a policy for moving through the graph so that the agent eventually reaches a desirable garage while balancing travel time, walking distance, and parking price.

In the implementation, the state is represented by the current node index in the graph, and the available actions correspond to moving to neighboring nodes. The Q-table stores an estimated value for each state-action pair, where the action is the destination node of an outgoing edge. The action-selection policy follows an ϵ -greedy strategy, with probability ϵ , the agent explores by selecting a random valid action, and otherwise it exploits by selecting the action with the highest current Q-value. The exploration rate is decreased linearly over training episodes.

The Q-values are updated using the standard Bellman update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s, a') - Q(s, a)]$$

where s is the current state, a is the selected action, s' is the next state, r is the immediate reward, α is the learning rate, and γ is the discount factor. If the agent reaches a terminal garage state, the future value term is set to zero.

The reward function is designed to encourage efficient progress through the graph while heavily favoring valid parking outcomes. For a non-terminal move, the reward consists of a travel-time penalty together with a distance-progress term:

$$R_{step} = -t(s, a) + 0.1[d(s, d) - d(a, d)]$$

where $t(s, a)$ is the current travel time of the selected edge, $d(s, d)$ is the Manhattan distance from the current node to the destination, and $d(a, d)$ is the Manhattan distance from the next node to the destination. This reward penalizes slow moves while giving a small advantage to actions that move the vehicle closer to the destination.

When the agent reaches a garage node, additional terminal rewards or penalties are applied. If the garage exceeds the maximum walking-distance threshold, a large negative penalty is assigned. If the garage is full, a smaller but still significant penalty is applied. If the garage is feasible, the agent receives

a large positive reward reduced by parking price and walking distance. The terminal reward for a valid garage can be written as

$$R_{garage} = -t(s, a) + 0.1[d(s, d) - d(a, d)] + 3000 - 3p_g - 20w_g$$

where p_g is the parking price and w_g is the walking distance from the garage to the destination. This reward structure allows the agent to learn both routing and parking behavior implicitly through interaction with the environment rather than through explicit garage preselection.

Training is performed over multiple episodes, with each episode beginning at the same start node and ending when a garage is reached or when a maximum number of steps is exceeded. During training, the environment evolves through state transitions, allowing the agent to learn under changing traffic and garage conditions. After training, the learned policy is evaluated by extracting the best path greedily from the Q-table until a garage is reached.

V. PERFORMANCE EVALUATION

The performance of the A* and Q-learning approaches was evaluated in the same dynamic urban parking environment using three main metrics, travel time, parking cost, and walking distance. These metrics were chosen because they directly reflect the objectives of the parking assignment problem, namely reducing the time spent driving, minimizing parking expense, and selecting a garage that remains reasonably close to the destination.

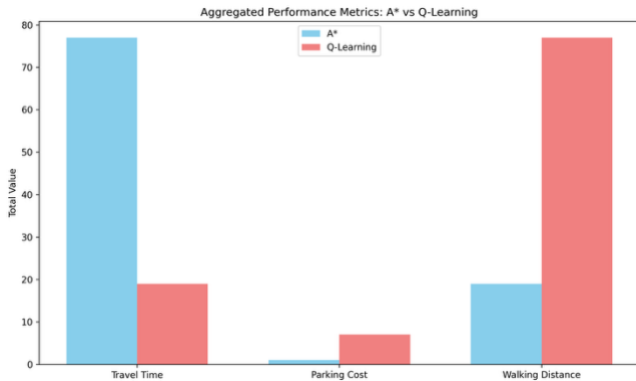


Fig. 3. Aggregated Performance Metrics

Fig. 3 presents the aggregated performance metrics for both approaches. The results show that the Q-learning approach achieved lower travel time, indicating that the learned policy was able to identify faster routing decisions in successful cases. In contrast, the A* approach achieved lower parking cost and shorter walking distance, showing that its explicit

garage-selection strategy was more effective at choosing cost-efficient garages near the destination. These results highlight a trade-off between the two methods: Q-learning performed better in reducing driving time, while A* performed better in selecting parking options that were cheaper and more convenient for the final walking stage.

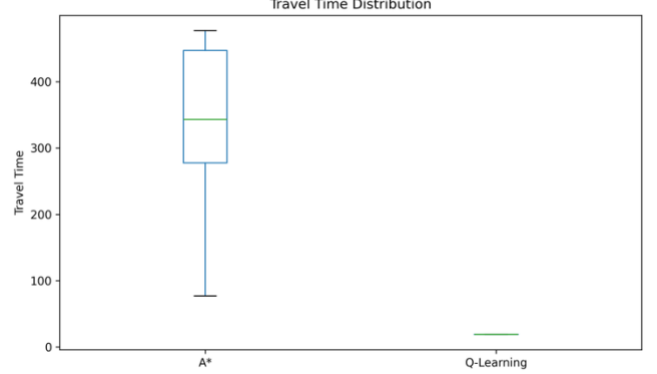


Fig. 4. Travel Time Distribution

Fig. 4 illustrates the distribution of travel times across successful trials. The A* approach shows a wider spread of values, indicating that its travel time varied more across different simulations. The Q-learning results are concentrated at a lower travel-time range, suggesting that when the learned policy succeeds, it can produce efficient routing outcomes. At the same time, the limited spread of Q-learning also reflects the fact that its performance was less stable overall, since some simulations did not lead to valid parking outcomes. This is consistent with the observed simulation failures for Q-learning, whereas A* remained more reliable under the tested conditions.

Overall, the results suggest that both approaches have value for the parking assignment problem, but they emphasize different strengths. A* provides a more dependable and interpretable baseline, with strong performance in parking cost and walking distance. Q-learning, on the other hand, shows promising potential for adaptive routing and lower travel time, although its reliability would likely benefit from further reward tuning and additional training. Under the tested conditions, A* offers the more robust overall solution, while Q-learning demonstrates the potential of learning-based methods for dynamic parking decision-making.

VI. CONCLUSION

This project studied the Dynamic Parking Assignment in Urban Areas problem as an AI-based decision problem over

an urban graph. The system integrates route planning and parking selection by considering travel time, parking price, garage capacity, and walking distance to the destination. Two methods were implemented: an A*- based heuristic search method first and now a Q-learning method.

The A* solution offers a strong baseline because it is computationally efficient, easy to interpret, and can be adapted to dynamic traffic through periodic rerouting. Its main weakness is that it depends on a fixed hand-crafted garage selection rule and does not improve from experience. By contrast, Q-learning provides a more flexible framework. Rather than explicitly computing the best garage using a formula, it learns action values through interaction with the environment. This gives it the potential to capture patterns in traffic and parking availability that may not be fully represented in a static cost model. This matches the broader direction of the current research, where reinforcement learning is increasingly used for sequential parking and traffic decisions under uncertainty [4][5].

Overall, the project demonstrates the DPAUA can be modelled effectively using AI methods and both search-based and learning-based approaches are viable. The A* method is preferable when fast, explainable decisions are needed, while Q-learning is more promising for future systems that must adapt over time. Future work should include more complete large-scale testing, richer traffic dynamic, better utility weighting between cost and walking distance, and possibly deep reinforcement learning for larger urban networks.

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